The Role of Temperature Gradients versus Static Stability on the Zonal Wind and Eddy Kinetic Energy Response to Thermal Perturbations

David J. Lorenz[a]

[a] *Center for Climatic Research, University of Wisconsin-Madison, Madison, Wisconsin*

Corresponding author: David J. Lorenz, dlorenz@wisc.edu

ABSTRACT: The relative role of meridional temperature gradients and static stability on the extratropical response to thermal perturbations are explored in a dry Primitive Equation (PE) GCM. A Quasi-Geostrophic (QG) model is used to separate the relative roles because the static stability in a QG model is externally prescribed and can therefore be altered independently of the meridional temperature gradient. For most experiments, the changes in meridional temperature gradients make the largest contribution to the zonal-mean zonal wind (U) changes, although the static stability changes are also important. In most cases, the mechanism for the static stability response does not directly involve the eddies. Instead, the advection of the static stability perturbation by the zonal-mean vertical velocity accounts for most of the U response to static stability. For increased static stability, advection by the Ferrel cell increases the meridional temperature gradient, which strengthens the jet via thermal wind. The stronger jet then shifts poleward, consistent with various theories in the literature. For Eddy Kinetic Energy (EKE), the direct effect of static stability makes a non-negligible contribution in most cases, however, the meridional temperature gradient and advection by Ferrel cell effects together are usually more dominant.

1. Introduction

The response of the atmosphere to idealized thermal perturbations (e.g. Haigh et al. (2005); Williams (2006); Lorenz and DeWeaver (2007); Frierson et al. (2007); Frierson (2008); Butler et al. (2010); McGraw and Barnes (2016); Wu and Smith (2016); Mbengue and Schneider (2017)) is a good starting point to understanding the response to increased greenhouse gases in comprehensive climate models. In this paper we focus on the response of the extratropical zonal-mean zonal winds (U) and Eddy Kinetic Energy (EKE) in a Primitive Equation (PE) and a Quasi-Geostrophic (QG) model on a sphere. The goal is to understand and quantify the relative role of meridional temperature gradients versus static stability (i.e. vertical temperature gradients) on the response to thermal perturbations. In particular, in the QG model, the static stability is prescribed a priori, so the temperature gradients and static stability can be changed independently to understand the total response.

The temperature response to increased greenhouse gases is not uniform, which gives rise to changes in horizontal and vertical temperature gradients. In comprehensive climate models, the tropical upper troposphere warms more than the extratropics due to the moist lapse rate constraint imposed by convection (Held (1993)). This increases the pole-to-equator temperature gradient and the baroclinicity leading to increases in EKE (Yin (2005); O'Gorman (2010)). By thermal wind, U also increases in the upper troposphere and stratosphere (Lorenz and DeWeaver (2007)). In addition, the dry static stability, which is synonymous with vertical temperature gradients, increases in the tropics due to the enhanced warming aloft. Dry static stability also tends to increase throughout the lower troposphere due to condensation from increased moisture. This increased static stability acts opposite the temperature gradients and contributes to decreases in EKE, especially in Northern Hemisphere summer (O'Gorman (2010)).

Increased greenhouse gases also cool the stratosphere, which leads to reduced static stability across the tropopause and, because of the downward tilt of the tropopause toward the poles, increased pole-to-equator temperature gradients. These temperature gradients reinforce those from the tropical upper troposphere leading to further increases in U at and above the tropopause (Lorenz and DeWeaver (2007)). The reduced static stability at the tropopause also leads to an upward shift in EKE (Yin (2005)).

In Northern Hemisphere winter, the low-level temperatures in the polar regions are projected to increase much more than the global average (Holland and Bitz (2003)). This polar amplification leads to a decrease in the pole-to-equator temperature gradient and a decrease in the static stability over the polar regions. The decreases in pole-to-equator temperature gradient by polar amplification act to decrease U (Butler et al. (2010)) and therefore oppose the increases discussed above.

So far, we have focused on changes in the magnitude of U and EKE, but greenhouse gases also cause latitudinal shifts in the circulation. Because the causes of these shifts are more subtle, prior studies have reached differing conclusions on the relative role of static stability versus meridional temperature gradients. For example, Frierson et al. (2006), Frierson (2008) and Lu et al. (2008) argue that the increased static stability due to global warming causes the poleward shift of the jet (U) and storm tracks (EKE): increased static stability causes the subtropical boundary between baroclinically stable/unstable to shift poleward thereby shifting the baroclinic wave generation and eddy momentum flux convergence poleward. Poleward shifts in the subtropical edge of the Hadley circulation have also been attributed to static stability via the same mechanism (Frierson et al. (2007); Kang and Lu (2012); Chemke and Polvani (2019)), however, the eddy driven jet and Hadley cell edge are not necessarily closely related for forced changes (Kang and Polvani (2011)). Other studies also believe static stability is important, but for different reasons. For example, Kidston et al. (2010) and Kidston et al. (2011) argue that increased static stability increases the horizontal length scale of the eddies which then shifts the jet poleward via interactions with the critical level on the poleward flank of the jet. Interestingly, all the static stability mechanisms above involve the effect of static stability directly on the eddies. The response of U then comes about via the eddy response. However, we find that the effect of static stability directly on the zonal mean is the larger effect (vertical advection of the vertical potential temperature gradients by the zonal-mean vertical velocity).

Other studies implicate meridional temperature gradients on the extratropical response. For example, Mbengue and Schneider (2017) find that the latitude of EKE maximum is primarily affected by surface temperature gradients. Harvey et al. (2014), Barnes and Screen (2015) and Harvey et al. (2015) discuss the response to thermal perturbations via their effect on meridional temperature gradients. Changes in meridional temperature gradients are also the essential component for dynamical mechanisms that invoke the paradigm that stronger jets shift poleward (Chen et al. (2007);

Kidston and Vallis (2012); Lorenz (2014a); Lorenz (2014b)). For example, Lorenz (2014b) show that there exists a selective "reflecting level" (i.e. turning latitude) on poleward flank of jet: for a given wavenumber, low phase speed waves are reflected but high phase speed waves are absorbed at the critical level on the poleward flank of jet. When the zonal-mean zonal wind increases, the peak of the reflecting level extends to higher phase speeds and a wider range of poleward propagating waves encounter a reflecting level instead of a critical level on the poleward flank. The increased wave reflection leads to increased equatorward propagating waves (and therefore poleward momentum flux) across the jet leading to a poleward shifted jet.

In this paper we perform simulations in the PE model with various thermal perturbations to the Held and Suarez (1994) forcing as in the following studies: Haigh et al. (2005), Williams (2006), Gerber and Vallis (2007), Lorenz and DeWeaver (2007), Butler et al. (2010) and McGraw and Barnes (2016). Next we perform analogous simulations in the QG model. The full response in the QG model is a combination of 1) perturbations to meridional temperature gradients from the Held and Suarez (1994) forcing and 2) the vertical static stability profile taken from the corresponding PE simulation. Next the two components are applied separately to quantify the relative contribution of meridional temperature gradients and static stability. The static stability response can be further decomposed into the contribution of vertical advection of the vertical potential temperature gradients by the zonal-mean vertical velocity and the effect of the static stability directly on the eddies.

The paper begins with a description of the PE and QG models. Next the mean state and response to thermal perturbations is given for the PE model. The the QG model is used to understand the relative contribution of the meridional temperature gradients and static stability in the PE simulations. Finally a suite of QG simulations with perturbed static stability at various vertical levels is analyzed to better understand the dynamics.

2. Methods

a. QG Model

Most QG models on a sphere assume a constant Coriolis parameter in both the geostrophic relationship and the stretching term in the QG Potential Vorticity (PV). Here, we instead use a QG

model on a sphere based on the linear balance equation (Whitaker and Loughe (1998)):

$$\nabla^2\left(\phi_j\right) = \nabla\cdot\left(f\nabla\left(\psi_j\right)\right), \tag{1}$$

where ψ is the streamfunction, f is the true Coriolis parameter, ϕ is the geopotential, and j is an index over pressure level. To be consistent with the literature of this model, j increases from the surface to the top of the atmosphere. The model is also formulated under the assumption of equally spaced pressure level (Δp), so that $p_j = 1000 + (j - \frac{1}{2})\Delta p$ mb. Lorenz (1960) developed the original model, but the vertical differencing is typically changed to a Charney-Phillips grid, and that is the version used here (Whitaker and Loughe (1998)). The vorticity equation is:

$$\frac{\partial\nabla^2\psi_j}{\partial t} + J\left(\psi_j, f + \nabla^2\psi_j\right) + \nabla\cdot\left(f\nabla\chi_j\right) = -k_M\nabla^2\psi_j - \nu\left|\nabla^{2n}\right|\nabla^2\psi_j, \tag{2}$$

where $J(\ldots)$ is the advection term written with the Jacobian (see below), χ is the velocity potential, k_M is the Rayleigh damping parameter, ν is the hyperdiffusivity parameter, and n is the order of the hyperdiffusion operator. The Jacobian is given by:

$$J\left(A, B\right) = \frac{1}{a^2}\frac{\partial A}{\partial\lambda}\frac{\partial B}{\partial\mu} - \frac{1}{a^2}\frac{\partial B}{\partial\lambda}\frac{\partial A}{\partial\mu}, \tag{3}$$

where a is the radius of the earth, λ is the longitude and μ is the sine of latitude. The vorticity equation (2) is the prognostic equation for the model, however, (2) includes the unknown variable χ. To close the system, one must consider the thermodynamic energy equation, which is given here in terms of the geopotential difference between levels:

$$\begin{aligned}\frac{\partial}{\partial t}\left(\phi_{j+1} - \phi_j\right) + \frac{1}{2}J\left(\psi_j + \psi_{j+1}, \phi_{j+1} - \phi_j\right) - \sigma_{j+1/2}\omega_{j+1/2} = \\ -k_T\left(\phi_{j+1} - \phi_j - \phi_E\right) - \nu\left|\nabla^{2n}\right|\left(\phi_{j+1} - \phi_j\right),\end{aligned} \tag{4}$$

where $\sigma_{j+1/2}$ is the QG static stability, $\omega_{j+1/2}$ is the vertical velocity, k_T is the thermal relaxation parameter and ϕ_E is the "radiative equilibrium" geopotential difference. All of these quantities are

defined on half levels. The velocity potential, χ, is related to the vertical velocity by

$$\nabla^2 \chi_j = \frac{\omega_{j+1/2} - \omega_{j-1/2}}{\Delta p}, \tag{5}$$

and with boundary conditions $\omega_{1/2} = 0$ and $\omega_{N+1/2} = 0$, where N is the number of vorticity levels. To eliminate the unknown variable χ in (2), take $\nabla \cdot \left(f \nabla \left(\nabla^{-2} \right) \right)$ of the difference of adjacent levels of (2) and ∇^2 of (4) and then use the balance equation, (1), to eliminate the time derivatives. This leads to a diagnostic "QG omega equation" for χ, which then is used to step the prognostic vorticity equation, (2), forward in time. For details see Whitaker and Loughe (1998).

The QG static stability is given by:

$$\sigma_{j+1/2} = c_p \left(\left(\frac{p_j}{p_0} \right)^{\kappa} - \left(\frac{p_{j+1}}{p_0} \right)^{\kappa} \right) \frac{\Theta_{j+2} + \Theta_{j+1} - \Theta_j - \Theta_{j-1}}{4 \Delta p}, \tag{6}$$

where c_p is the specific heat at constant pressure, $\kappa = 2/7$, and Θ is the horizontally uniform reference potential temperature. The unusual Θ differencing in the last factor is the effective $\Delta\Theta/\Delta p$ stencil for a PE model with second order differencing in the vertical. This form is also less sensitive to noise in Θ and is approximately equal to $\frac{\Theta_{j+1} - \Theta_j}{\Delta p}$ for smooth profiles. At the two boundaries, one should set either $\Theta_{j+2} = \Theta_{j+1}$ or $\Theta_j = \Theta_{j-1}$, which is also based on the PE differencing. In this paper, the reference Θ_j is taken from the pressure-level PE model, which is described below. The forcing in the QG model (k_m, k_T, ϕ_E) is from Held and Suarez (1994). The hyperdiffusion is 8th order with a time scale of 0.1 days for the smallest scale waves. Third-order Adams-Bashforth time differencing (Durran (1991)) is used. The resolution is T42 with 20 equally spaced pressure levels. For each experiment below, the model is run for 6500 days and the first 500 days are discarded for spin-up.

The QG approximation assumes that horizontal advection is by the non-divergent wind alone. Therefore, for a consistent the potential temperature and energy budget, the thermodynamic variables in the QG system must be partitioned into a reference state, which depends on pressure only, and the deviations from the references state, which depend on longitude, latitude, pressure, and time (White (2002)). In other words, the reference state static stability (or potential temperature) is prescribed and is independent of the evolution of the model. In this paper, this limitation of the QG approximation is used to isolate the effects of static stability from horizontal temperature gradients.

For example, in the thermodynamic equation, (4), $\sigma_{j+1/2}$ is the reference state and $\phi_{j+1} - \phi_j$ is the temperature deviation. σ, the static stability, is a constant that is not altered by heating or vertical advection. In addition, note that at each level the horizontal mean $\phi_{j+1} - \phi_j$ does not participate in the dynamics on the left-hand side[1], therefore only horizontal variations in ϕ matter. Therefore, the heating on the right-hand side in (4) only effects the horizontal gradients. In other words, when we change the Held and Suarez (1994) parameters in the QG model, we are only impacting the horizontal temperature gradients. To change the vertical gradients, we must separately change σ.

b. PE Model

Like the QG model, the PE model is also in pressure coordinates (Rivier et al. (2002)):

$$\begin{aligned}
\frac{\partial \eta}{\partial t} &= -\frac{1}{a(1-\mu^2)}\frac{\partial A}{\partial \lambda} - \frac{1}{a}\frac{\partial B}{\partial \mu}, \\
\frac{\partial \delta}{\partial t} &= -\frac{1}{a(1-\mu^2)}\frac{\partial B}{\partial \lambda} - \frac{1}{a}\frac{\partial A}{\partial \mu} - \nabla^2\phi - \nabla^2 E, \\
\frac{\partial \theta}{\partial t} &= -\frac{1}{a(1-\mu^2)}\frac{\partial C}{\partial \lambda} - \frac{1}{a}\frac{\partial C}{\partial \mu} - F,
\end{aligned} \tag{7}$$

where η is the absolute vorticity, δ is the divergence, θ is the potential temperature and

$$\begin{aligned}
A &= U\eta + V\delta + \frac{\partial V\omega}{\partial p}, \\
B &= V\eta - U\delta - \frac{\partial U\omega}{\partial p}, \\
C &= U\theta, \\
D &= V\theta, \\
E &= \frac{U^2+V^2}{2(1-\mu^2)}, \\
F &= \frac{\partial \omega\theta'}{\partial p},
\end{aligned} \tag{8}$$

where U and V are $u\cos(\text{latitude})$ and $v\cos(\text{latitude})$, respectively, and θ' is the deviation from the reference, isothermal θ used for the semi-implicit method. The upper and lower boundary conditions are $\omega = 0$, which implies that the net divergence in a vertical column is zero. To force

[1]As written, the horizontal-mean $\phi_{j+1} - \phi_j$ in (4) will exponentially decay to the horizontal-mean of ϕ_E, but this has no impact on horizontal variations in $\phi_{j+1} - \phi_j$.

this constraint, we use the divergence difference, $\hat{\delta}_{j+1/2} = \delta_{j+1} - \delta_j$, at $n-1$ levels as a prognostic variable (where n is the number of vertical levels). The full divergence, which is required in (8), is calculated from the $\hat{\delta}$ definition and the contraint that the mass-weighted vertical integral of δ is zero. Because of the constant surface pressure approximation, one might expect the near surface winds to be constrained close to zero. However, because the divergence *difference* is the prognostic variable, only the geopotential *difference* between levels enters the dynamical equations. The geopotential difference between levels is proportional to the temperature just like a sigma coordinate GCM, so the evolution of the divergence difference is unaffected. The constant surface pressure approximation only enters the dynamics via the constraint that the vertically integrated divergence is zero. Details of the vertical differencing are found in Rivier et al. (2002). Like the QG model, the resolution is T42 with 20 equally spaced pressure levels. The model is forced with the diabatic heating and frictional damping of Held and Suarez (1994). An 8th order hyperdiffusion with a time scale of 0.1 days for the smallest scale waves is added to the three pronostic variables. The time differencing uses the semi-implicit AB3-AI2 method of Durran and Blossey (2012). Like the QG model, the PE model is run for 6500 days and the first 500 days are discarded for spin-up.

Despite the constant surface pressure assumption, the circulation of the pressure coordinate model appears to be unaffected. For example, we have compared the model to sigma coordinate models with two separate vertical differencing schemes (Hoskins and Simmons (1975) and Simmons and Burridge (1981)). At low resolution (T42), the jet latitude and persistence of the sigma coordinate models are farther apart from each other than either is to the pressure coordinate model (i.e. the pressure coordinate model is the median model). At higher resolution, all models converge to the same circulation. We are also aware of the potential impact of constant surface pressure on the transient response of the zonal-mean zonal wind to forcing and heating (Haynes and Shepherd (1989)), however, the EOF patterns appear to be essentially unaffected.

TABLE 1. Perturbation experiments with Held and Suarez (1994) GCM.

Name	Description	Control Value	Perturbed Value
TSTRAT	decrease stratosphere temp.	T_{strat} = 200 K	T_{strat} = 190 K
$\Delta\theta_z$TROP	increase tropical static stability	$\Delta\theta_z$ = 10 K	$\Delta\theta_z$ = 15 K
ΔT_y	increase pole-to-eq. ΔT but keep tropopause height at 45° same	ΔT_y = 60 K, $T_{\max}$ = 315 K	ΔT_y = 75 K, $T_{\max}$ = 322.5 K
POLE	heating at low-levels over pole (polar amplification)	none	see text

c. Perturbation Experiments

The Held and Suarez (1994) heating used in our models acts to drive the temperature to a prescribed "equilibrium" temperature profile that depends on pressure, p, and latitude, ϕ:

$$T_{eq} = \max\left(T_{strat}, \left[T_{max} - (\Delta T)_y \sin^2\phi - (\Delta\theta)_z \log\left(\frac{p}{p_0}\right)\cos^2\phi\right]\left(\frac{p}{p_0}\right)^{\kappa}\right), \tag{9}$$

where T_{strat} is the stratosphere temperature, $(\Delta T)_y$ is the pole-to-equator temperature gradient, $(\Delta\theta)_z$ is a static stability constant, $p_0 = 1000$mb, and $\kappa = 2/7$. For the first perturbation experiment, we reduce T_{strat} from the control Held and Suarez (1994) value of 200K to 190K (see line 1 of Table 1). Decreasing T_{strat} means the tropospheric lapse in T_{eq} extends higher in the atmosphere and so this experiment is equivalent to an increase in the height of the tropopause (Haigh et al. (2005); Williams (2006)). For the second perturbation experiment, the static stability parameter, $(\Delta\theta)_z$, is increased from 10K to 15K. Because this term is multiplied by $\cos^2\phi$ in (9), this change primarily affects the static stability in the tropics (see line 2 of Table 1)). For the third experiment we explore changes in the pole-to-equator temperature gradient, $(\Delta T)_y$. To minimize the impact on tropopause height, we change both $(\Delta T)_y$ and T_{max} such that the tropopause height at 45°, which is approximately the location of the mid-latitude jet, is unchanged (see line 3 of Table 1). For the final experiment, we apply cooling at low-levels over the polar regions to emulate polar amplification under increased greenhouse gases. The Held and Suarez (1994) forcing cannot be altered to achieve this cooling, therefore we add the following change to T_{eq} for polar amplification:

$$\Delta T_{eq} = \begin{cases} 40K \cdot \frac{p-600\text{mb}}{p_0-600\text{mb}}, & \phi > 65^\circ \text{ and } p > 600\text{mb} \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

d. Localized Static Stability Experiments

We also perform a set of QG experiments with localized static stability changes. For each experiment the σ perturbation is the form of a Gaussian function:

$$\Delta\sigma_{j+1/2} = c_p \left(\left(\frac{p_j}{p_0} \right)^{\kappa} - \left(\frac{p_{j+1}}{p_0} \right)^{\kappa} \right) \frac{A \exp\left(\left(\frac{p_{j+1/2} - p_c}{w} \right)^2 \right)}{\Delta p}, \tag{11}$$

where the A is 2K, w is 100mb and p_c is the location of the perturbation. We perform ten simulations with different p_c ranging from 50mb to 950mb in increments of 100mb. Recall that for consistent energetics, the static stability of a QG model is constant in the horizontal so the above perturbations depend on pressure only.

3. Results

a. Mean State

The time-mean state in the PE and QG model are shown in Fig. 1 and 2, respectively. For the zonal-mean zonal wind (U), the mid-latitude jet is located around 42° in the PE model. In the QG model, the jet is shifted poleward (47°) and the "shoulder" of stronger subtropical U is more prominent. The near surface U is also stronger in the QG model. For eddy kinetic energy (EKE, panel b), the QG field is also further poleward, but the amplitude is slightly weaker. For the eddy momentum flux ($\overline{u'v'}$, panel c), the amplitude of the QG field is significantly stronger than for the PE model, which is consistent with the stronger near surface U in the QG model. The weaker $\overline{u'v'}$ but stronger EKE of the PE model suggests that the PE jet is more like a waveguide than the QG jet (waveguide: waves are trapped in latitude due to reflection on the jet flanks). Also, consistent with the waveguide hypothesis is the fact that the vertical EP flux (panel d), or baroclinic wave source, is weaker in the PE model yet the EKE is stronger. So in other words, the waves that are created in the PE model linger in the mid-latitudes longer, whereas in the QG model they have a greater tendency to propagate to the subtropics where they dissipate at the critical line. We believe this waveguide issue is very interesting and it appears that there is more to the problem than simply the jet latitude, but we currently do not have a good understanding of this issue. Therefore, this is not discussed further in this paper.

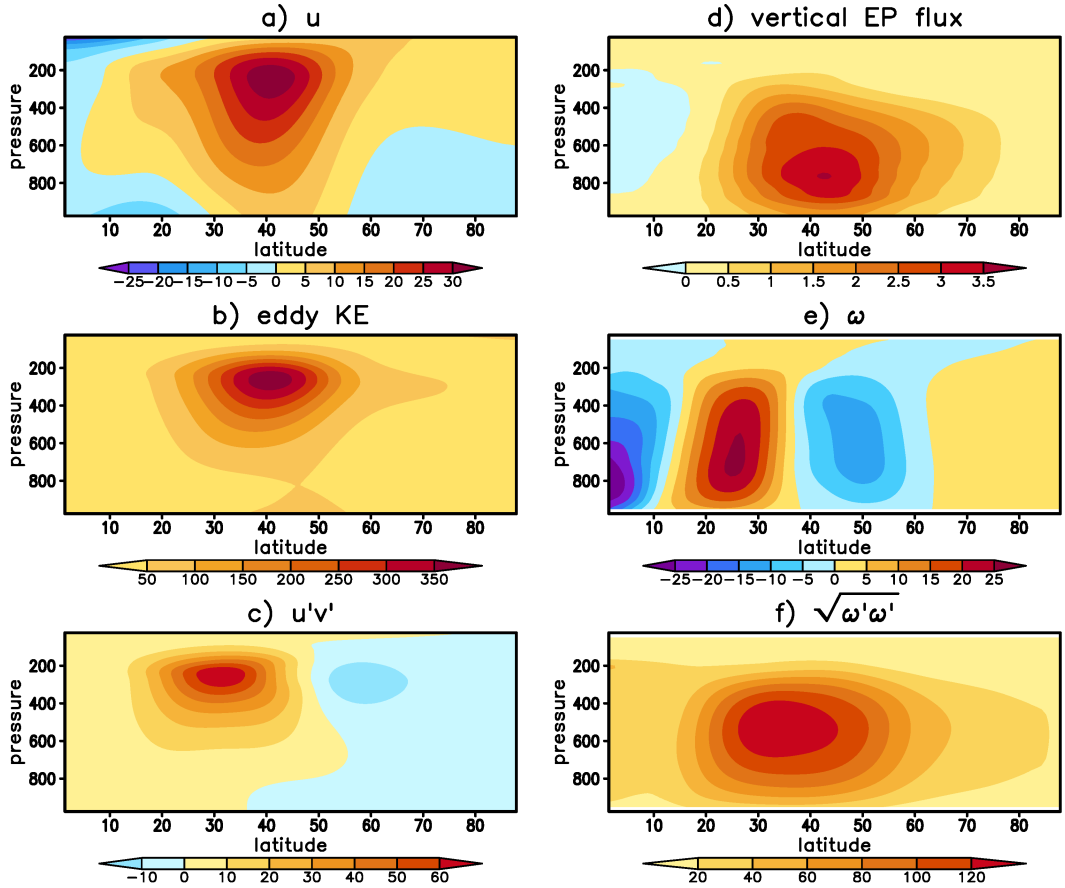

FIG. 1. Time-mean fields from PE GCM. a) zonal-mean zonal wind (m/s). b) zonal-mean eddy kinetic energy (m^2/s^2). c) zonal-mean eddy momentum flux (m^2/s^2). d) vertical component of the EP flux divided by radius of earth ($Pa \cdot m/s^2$). e) zonal-mean pressure vertical velocity (ω, mPa/s). f) square root of the zonal-mean of the eddy ω variance (mPa/s).

Finally, we show vertical velocity ($= \omega$) statistics since this field is important for understanding the results below. In the tropics and subtropics, the zonal-mean ω circulation is stronger in the PE model (panel e). In the mid-latitudes, on the other hand, the ω circulation is about the same. For the eddy standard deviation (panel f), the QG model field is larger.

b. Response to Thermal Forcings

The responses of U and EKE in the PE model for the experiments in Table 1 are shown in Fig. 3. As shown in Haigh et al. (2005), Williams (2006) and Lorenz and DeWeaver (2007), decreasing the Held and Suarez (1994) stratosphere temperature raises the tropopause and causes a poleward

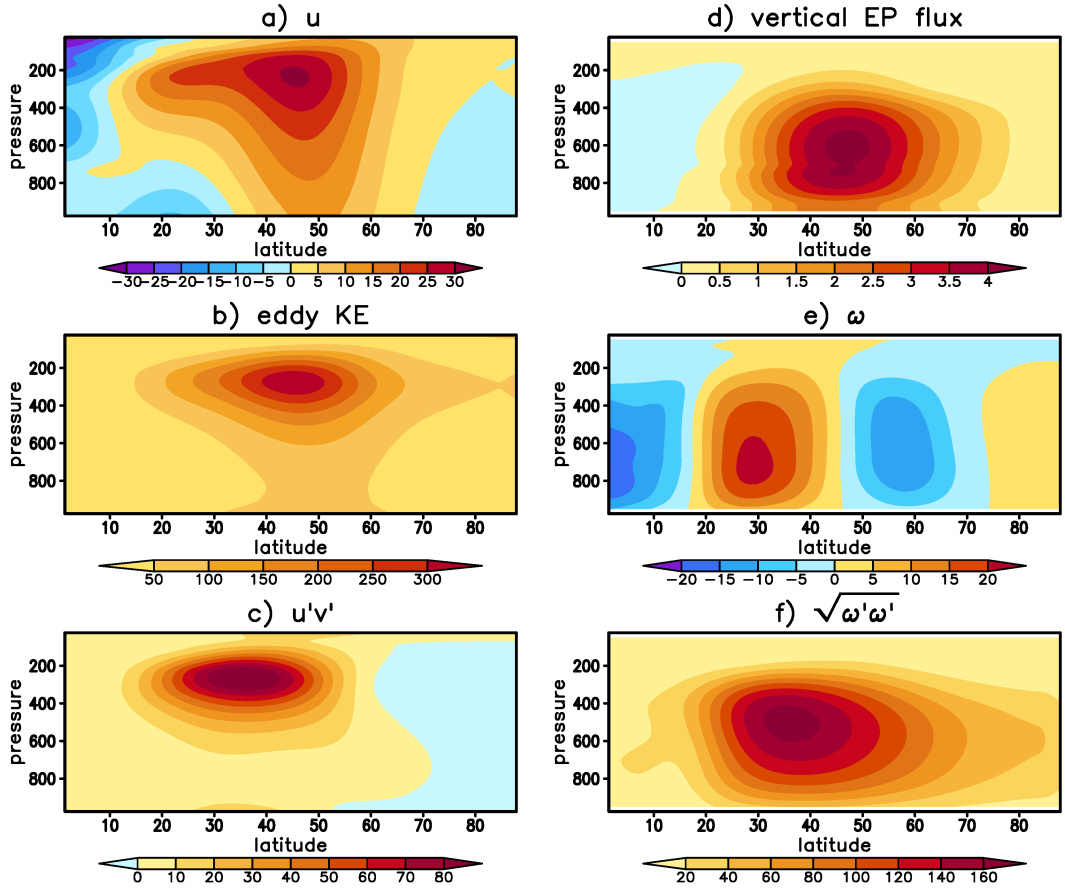

FIG. 2. Same as Fig. 1 but for the QG model.

shift the mid-latitude jet. The EKE also shifts poleward and upward in the TSTRAT simulation. For the $\Delta\theta_z$TROP experiment the jet also shifts poleward (Butler et al. (2010); McGraw and Barnes (2016)). The U changes look similar to TSTRAT except the U changes equatorward of 40° are larger in amplitude. For the EKE, the decreases in EKE on the equatorward flank of the jet are much more pronounced compared to TSTRAT and there is much less upward shift. For the ΔT_y experiment (e.g. Gerber and Vallis (2007)), the U changes on the poleward flank of the jet are more pronounced compared to the previous experiments and the EKE response is a monopole (Fig. 3g), unlike other simulations. Unlike the other experiments, polar amplification (POLE) causes an equatorward shift of U and EKE (Butler et al. (2010)).

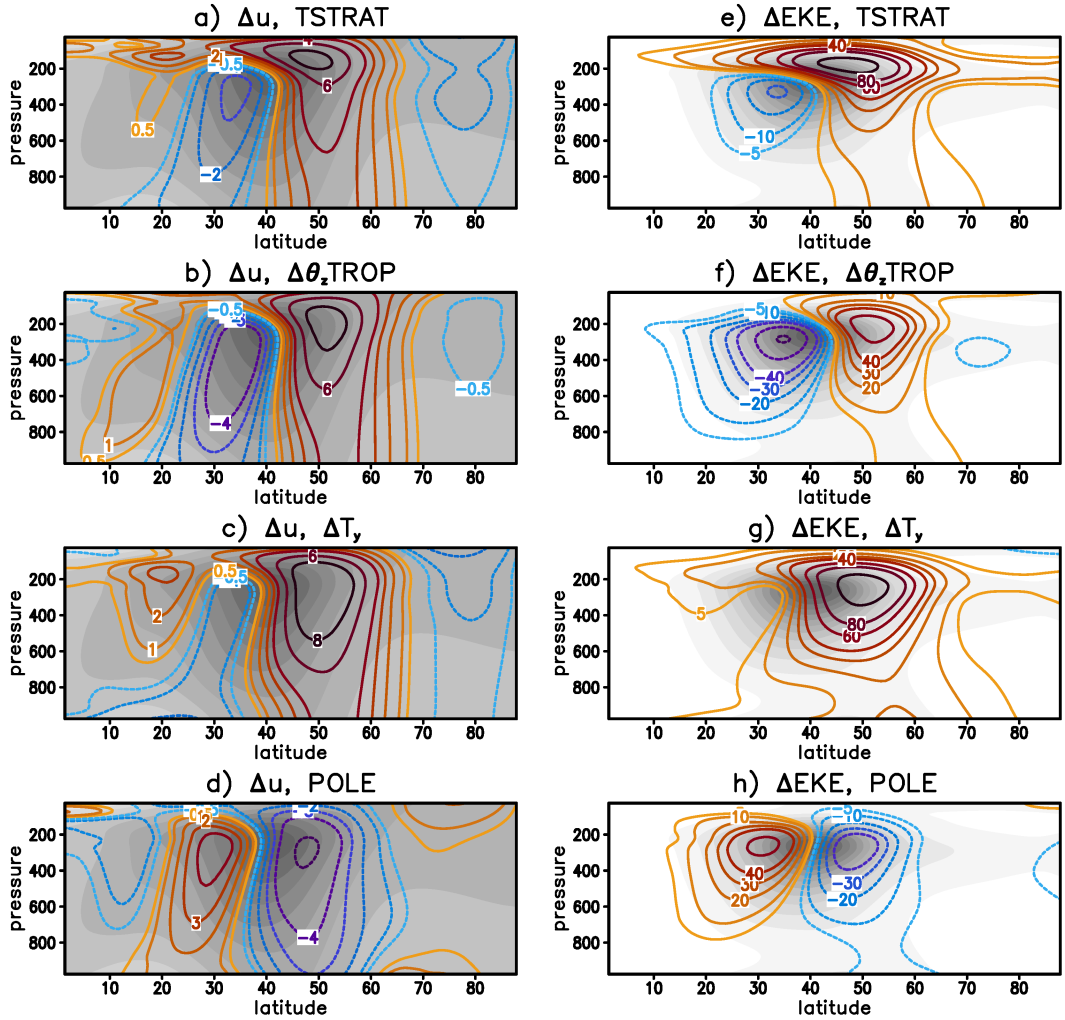

FIG. 3. PE results: a) Change in zonal-mean zonal wind for the TSTRAT experiment (see Table 1). b) Same as (a) but for the $\Delta\theta_z$TROP experiment. c) Same as (a) but for the ΔT_y experiment. d) Same as (a) but for the POLE experiment. e) Change in eddy kinetic energy for the TSTRAT experiment. f) Same as (e) but for the $\Delta\theta_z$TROP experiment. g) Same as (e) but for the ΔT_y experiment. h) Same as (e) but for the POLE experiment.

c. Temperature Gradients versus Static Stability

In this subsection, we look at the change experiments (Table 1) in the QG model and we diagnose the relative role of meridional temperature gradients ($\partial\theta/\partial y$) and static stability ($d\Theta/dp$) for the U and EKE changes. Like any QG model, the static stability is prescribed, therefore to best represent the PE model, the QG experiments must change both the Held and Suarez (1994) parameter(s) and the static stability, which is taken from the appropriate PE simulation. For TSTRAT (Fig. 4), the

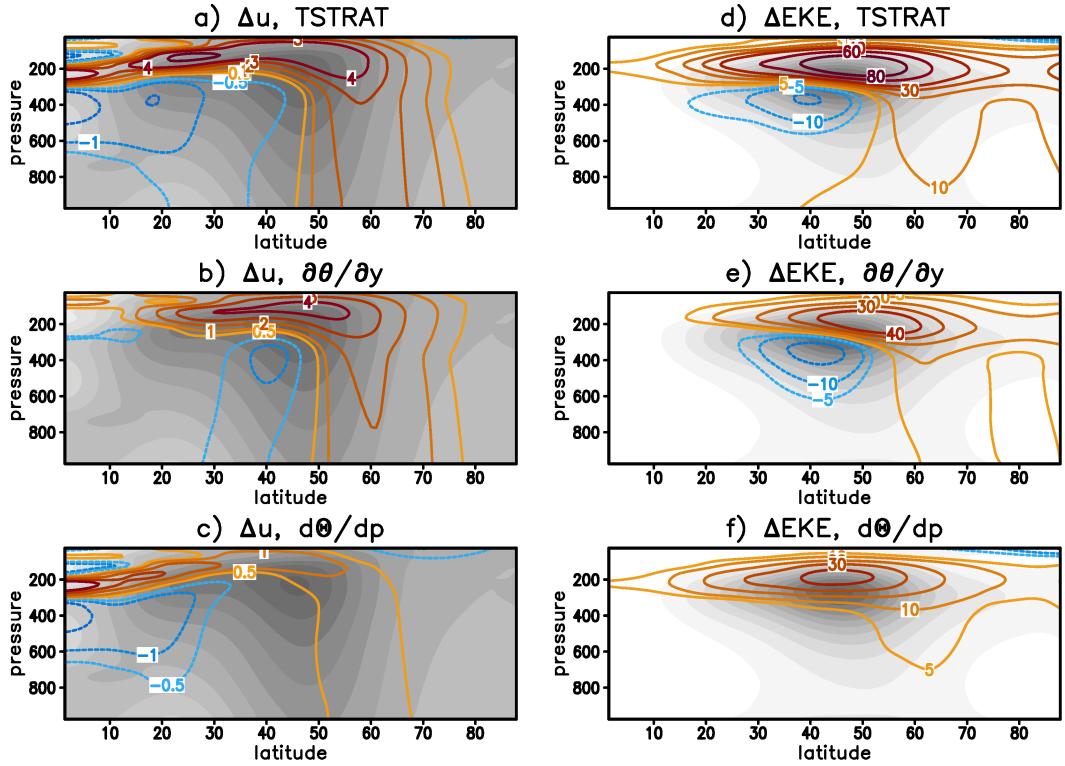

FIG. 4. QG model results: a) Change in zonal-mean zonal wind for the TSTRAT experiment (see Table 1). b) As in (a) but for the meridional temperature gradient forcing from TSTRAT and the static stability from the control simulation. c) As in (a) but for the meridional temperature gradient forcing from the control simulation and the static stability from TSTRAT. d) As in (a) but for eddy kinetic energy. e) As in (b) but for eddy kinetic energy. f) As in (c) but for eddy kinetic energy.

QG U change shares the general features of the PE model change: U increases above and on the poleward flank of the jet and there are weaker negative anomalies on the equatorward flank. The amplitude of the QG response is weaker, which is expected given the reduced persistence of the leading U EOF in the QG model (not shown). Changing $\partial\theta/\partial y$ only (i.e. change the Held and Suarez (1994) parameters only), captures most of the U response in the stratosphere and much of the U response in the poleward center of action in the troposphere (Fig. 4b). The $d\Theta/dp$ changes account for a smaller portion of the poleward center of action, but most of the U decreases in the tropics (Fig. 4c). These tropical changes, however, are not present in the PE simulation (Fig. 3a).

The relative roles of $\partial\theta/\partial y$ and $d\Theta/dp$ on the lower tropospheric average (1000-700mb) U are summarized in Table 2. The "slope" metric gives the regression coefficient, b, that minimizes

TABLE 2. Mechanisms of the response of zonal-mean zonal wind averaged vertically from 1000mb to 700mb (see text).

	$\partial\theta/\partial y$		$d\Theta/dp$		$\partial\theta/\partial y$ plus $\overline{\omega}$ on $d\Theta/dp$		$d\Theta/dp$ minus $\overline{\omega}$ on $d\Theta/dp$	
simulation	slope	corr	slope	corr	slope	corr	slope	corr
TSTRAT	0.74	0.97	0.29	0.93	0.34	0.88	0.78	0.96
$\Delta\theta_z$TROP	0.33	0.62	0.56	0.82	1.11	0.97	-0.30	-0.73
ΔT_y	0.70	0.85	0.20	0.50	0.95	0.98	-0.02	-0.13
POLE	0.57	0.93	0.36	0.82	0.99	0.99	0.02	0.10

the difference between the full response and the partial (i.e. either $\partial\theta/\partial y$ or $d\Theta/dp$) response: $\min\left(U_{\text{Full}} - sU_{\text{Partial}}\right)^2$. The minimization is over latitude only, since U is averaged vertically first. An s close to one means the partial response captures the correct amplitude and therefore the corresponding mechanism accounts for the full response. Lower values mean the alternative mechanism is important. The "corr" metric gives the spatial correlation between U_{Full} and U_{Partial}. Low values of this metric can detect cases with "proper" amplitude but "wrong" spatial pattern (i.e. correct amplitude but for the wrong reason). In summary, a partial response does not account for the full response unless *both* the slope and correlation are close to one.

For the TSTRAT experiment, we see that the spatial correlation is large for both temperature gradients ($\partial\theta/\partial y$) and static stability ($d\Theta/dp$) but that the temperature gradients account for about 3/4 of the total response. The relative role of these two processes was also explored in Lorenz and DeWeaver (2007) using a large number of imposed thermal perturbations, however, the conclusions of that analysis were not clear. In a QG model, on the other hand, the temperature gradient and static stability are separate and so a clear quantitative assessment can be made. Finally, when the U response in Fig. 4b and c are added to together we find that the response is very close to Fig. 4a and therefore the linear superposition principle holds (not shown). Superposition also holds for both U and EKE for all remaining experiments discussed below.

The EKE change in the QG model is similar in structure, but the tropospheric anomalies are slightly too weak (Fig. 4d). This is consistent with the weaker tropospheric U response. In the stratosphere the EKE responses to $\partial\theta/\partial y$ and $d\Theta/dp$ are similar in amplitude, in the troposphere, however, only the $\partial\theta/\partial y$ simulation captures the negative EKE anomalies below and equatorward of the time-mean EKE. To quantify the relative contribution of these two mechanisms to the total

TABLE 3. Mechanisms of the response of eddy kinetic energy (see text).

	$\partial\theta/\partial y$		$d\Theta/dp$		$\partial\theta/\partial y$ plus $\overline{\omega}$ on $d\Theta/dp$		$d\Theta/dp$ minus $\overline{\omega}$ on $d\Theta/dp$	
simulation	slope	corr	slope	corr	slope	corr	slope	corr
TSTRAT	0.48	0.94	0.43	0.93	0.23	0.71	0.75	0.95
$\Delta\theta_z$TROP	0.41	0.42	0.50	0.51	0.93	0.62	-0.05	-0.04
ΔT_y	1.05	0.94	-0.05	-0.18	1.15	0.96	-0.13	-0.47
POLE	0.74	0.93	0.23	0.56	0.99	0.97	0.03	0.09

response see Table 3. The EKE metrics are the same as U (Table 2), except the regressions and correlation are over latitude and pressure instead of only latitude. For TSTRAT, the spatial correlations are similar and $\partial\theta/\partial y$ accounts for slightly more of the total response.

In the next perturbation experiment ($\Delta\theta_z$TROP), the tropical static stability parameter in Held and Suarez (1994) is increased from 10K to 15K. The U response in the QG model (Fig. 5a) is broadly similar to the PE model (Fig. 3b), although the amplitude is weaker as before. In this experiment, the response to $d\Theta/dp$ (Fig. 5c) is closer to the full response than $\partial\theta/\partial y$ (Fig. 5b). For example, the former shows a dipole U structure in the mid-latitudes while the latter does not. This is reflected in the slope metrics in table 2: $d\Theta/dp = 0.56$ and $\partial\theta/\partial y = 0.33$. The maximum U amplitude, however, is comparable in the two experiments.

The poleward shift with increased static stability seems consistent with the mechanism in Frierson (2008) and Lu et al. (2008). Even in this QG model, with horizontally uniform static stability, there exists a boundary in the subtropics between baroclinic stable and unstable latitudes due to the varying Coriolis parameter. As the static stability increases, the boundary moves poleward, which moves baroclinic instability and the eddy driven jet poleward (Frierson (2008); Lu et al. (2008)). Looking deeper, however, we find that the response is not due to the impact of $d\Theta/dp$ on the waves but on the zonal-mean.

The impact of $d\Theta/dp$ on the zonal-mean occurs via A) the heating associated with the control run $\overline{\omega}$ acting on the change in static stability ($\Delta d\Theta/dp$ for $\Delta\theta_z$TROP minus control) and B) the heating and momentum forcing associated with the static-stability-induced change in the mean meridional circulation ($\Delta\overline{v}$ and $\Delta\overline{\omega}$) acting on the control $d\Theta/dp$ and the Coriolis parameter. We find that mechanism B is negligible because almost all the mean meridional circulation changes in the extratropics are coming from the eddies and not the static stability. For example, Fig. 6

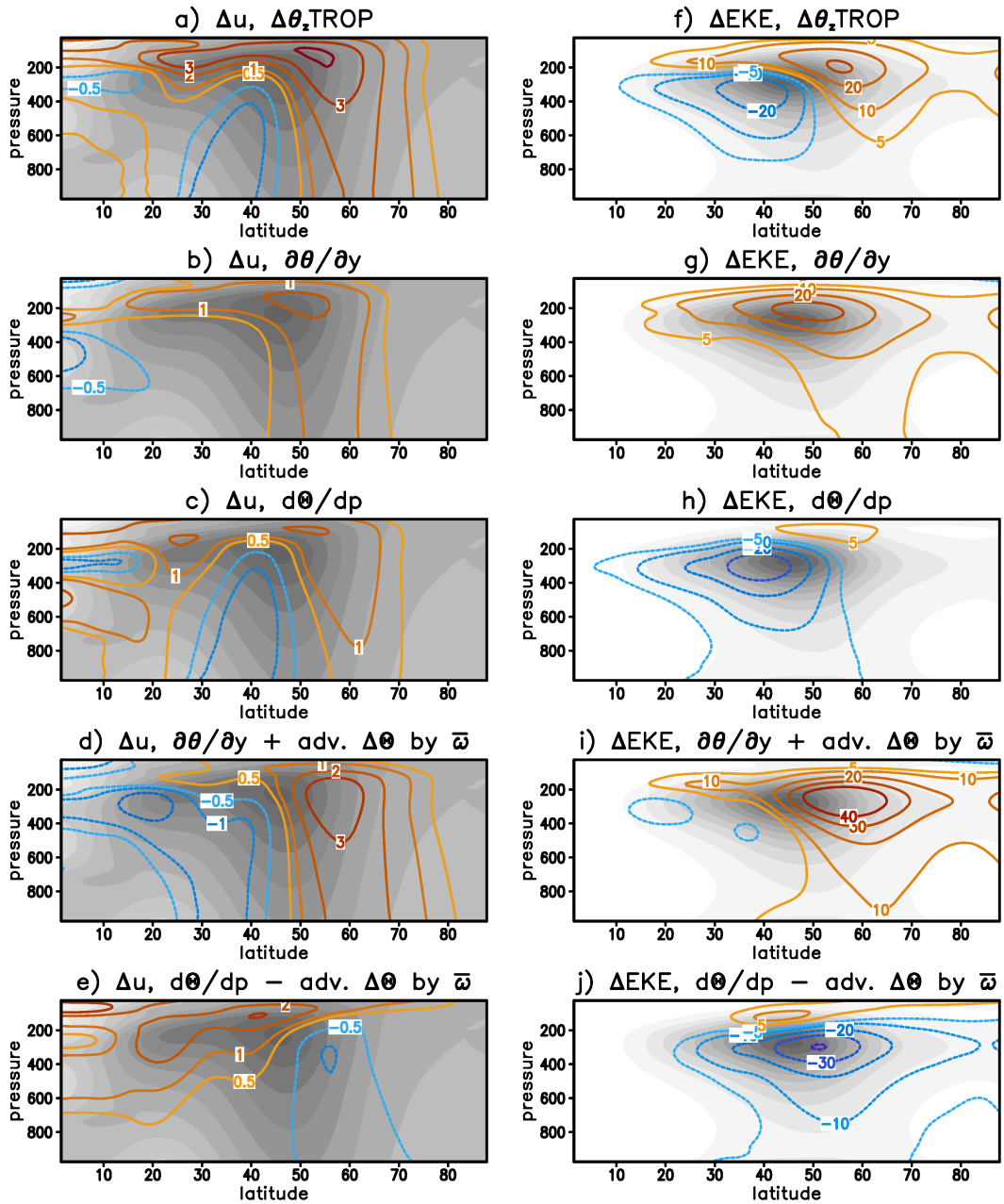

FIG. 5. QG model results: a) Change in zonal-mean zonal wind for the $\Delta\theta_z$TROP experiment (see Table 1) (m/s). b) As in (a) but for the meridional temperature gradient forcing from $\Delta\theta_z$TROP and the static stability from the control simulation. c) As in (a) but for the meridional temperature gradient forcing from the control simulation and the static stability from $\Delta\theta_z$TROP. d) As in (b) but add heat source equal to zonal-mean ω acting on change in static stability from $\Delta\theta_z$TROP (i.e. vertical advection of Θ by $\overline{\omega}$). e) As in (c) but subtract heat source equal to zonal-mean ω acting on change in static stability from $\Delta\theta_z$TROP. (f-j) As in (a-e) but for EKE ($m^2 s^{-2}$).

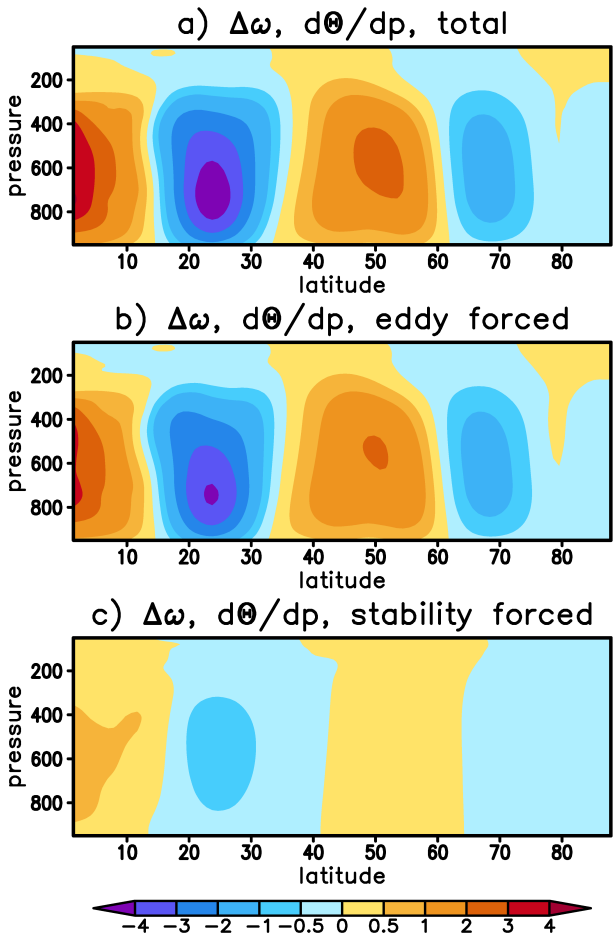

FIG. 6. a) Response of zonal-mean vertical pressure velocity (ω) to the change in the static stability from $\Delta\theta_z$TROP. b) Same as (a) but for the contribution of changes in the eddy fluxes to the $\overline{\omega}$. c) Same as (a) but for the contribution of changes in the static stability acting directly on $\overline{\omega}$ with no changes in eddy fluxes.

shows the total $\overline{\omega}$ response from $d\Theta/dp$ as well as the contribution of the eddies and static stability changes. The contribution of the static-stability-induced mean meridional circulation is small enough that it has a negligible impact on the heat and momentum budget via mechanism B (not shown). The smallness to the static stability induced circulation is even more pronounced in the other perturbation experiments (not shown). Therefore, the impact of $d\Theta/dp$ on the zonal-mean occurs via mechanism A.

In the experiment shown in Fig. 5d, the heating associated with mechanism A is added to the Held and Suarez (1994) physics together with the $\partial\theta/\partial y$ from $\Delta\theta_z$TROP. The static stability, however, remains fixed at the control run values. The U response in this new simulation (Fig. 5d)

attains the amplitude of the full simulation in Fig. 5a. This is also documented in Table 2: the spatial correlation and slope are 0.97 and 1.11. Therefore, the direct impact of static stability on the eddies is not responsible for the poleward shift in this case. Instead, the mechanism can be seen by looking at the time-mean $\overline{\omega}$ in Fig. 2e, which has downward motion at 28° and upward motion at 57°. When the static stability increases, $\overline{\omega}$ leads to heating at 28° and cooling at 57°, which increases the temperature gradient across the jet. This strengthens the jet and causes a poleward shift consistent with the mechanism in Kidston and Vallis (2012), Lorenz (2014a) and Lorenz (2014b) (see Introduction).

Nevertheless, the direct impact of static stability on the eddies is not zero. To better see this effect we perform the complement of the experiment in Fig. 5d: we change $d\Theta/dp$ and then *subtract* the heating associated with the control run $\overline{\omega}$ acting on the change in $\Delta d\Theta/dp$ (Fig. 5e). As before, linear super-position holds (not shown). U increases at upper-levels equatorward of the mean jet and decreases slightly on the polar flank of the jet. This direct eddy response to static stability opposes the full response. This U response appears related to the accompanying EKE decrease in Fig. 5j. The decreases in EKE are associated with decreases in the vertical and horizontal wave activity, or EP, fluxes (not shown), which imply a net positive U force at upper-levels equatorward of the jet (where wave dissipation typically occurs) and a net negative U force at the latitudes of the negative EKE response (just poleward of the jet core in this case).

The EKE response in the QG model (Fig. 5f) is weaker than the PE model (Fig. 3f), which is consistent with the weaker U response. The role of $\partial\theta/\partial y$ versus $d\Theta/dp$ is dramatically different with $\partial\theta/\partial y$ responsible for the increased EKE (Fig. 5g) and $d\Theta/dp$ responsible for the decreased EKE (Fig. 5h). Adding the Θ advection by $\overline{\omega}$ to the $\partial\theta/\partial y$ effect (Fig. 5i) increases the EKE even more except for some weak decreases in the subtropics. Evidently, the decreases in EKE are due to the direct impact of static stability on the waves (increased static stability implies decreased baroclinicity), as demonstrated in Fig. 5j. This is in contrast to U, where $\partial\theta/\partial y$ and Θ advection by $\overline{\omega}$ explain most of the total response.

The sensitivity of EKE to the details of the zonal-mean temperature response has been reported by Yuval and Kaspi (2020). They find that as the temperature response becomes broader, the meridional temperature gradients become less important and the static stability changes become more important. Our experiment with changes in static stability alone (Fig. 5h), which is equivalent

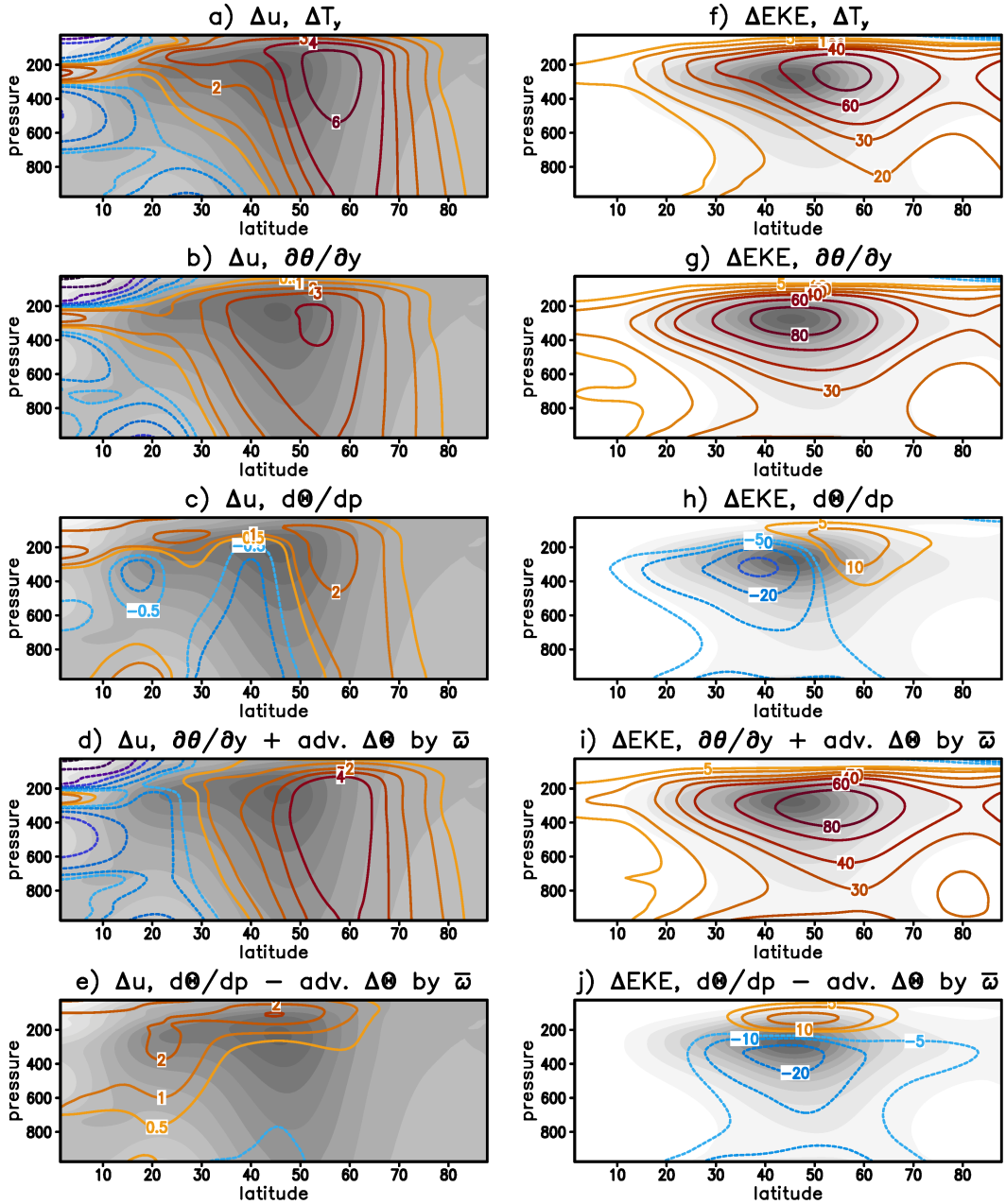

FIG. 7. As in Fig. 5 but for the ΔT_y experiment (see Table 1).

to the broadest possible temperature response, confirm that the EKE weakens for broad temperature changes. U, on the other hand, shifts poleward for both kinds of temperature responses because static stability changes impact the meridional temperature gradient via advection by $\overline{\omega}$. This is also consistent with Yuval and Kaspi (2020) (see their figure 4).

In the next perturbation experiment (ΔT_y), the pole-to-equator temperature gradient is increased (Table 1). Like the $\Delta\theta_z$TROP, 1) the $\partial\theta/\partial y$ response is mostly a monopole in the mid-latitudes (Fig. 7b) while the $d\Theta/dp$ response is clearly a dipole (Fig. 7c) and 2) adding the Θ advection by $\overline{\omega}$ to the $\partial\theta/\partial y$ response, we recover essentially the full U response at low levels (Fig. 7d). The effect of static stability acting directly on the eddies (Fig. 7e) is very weak except for upper-levels at and equatorward of the mean jet.

For EKE, the response is somewhat similar to $\Delta\theta_z$TROP in that the $\partial\theta/\partial y$ response is dominated by EKE increases and the $d\Theta/dp$ response has more prominent EKE decreases (Fig. 7gh). Also, like the $\Delta\theta_z$TROP, adding Θ advection by $\overline{\omega}$ improves the representation of the total response on the poleward flank of the storm track, but not so much on the equatorward flank (Fig. 7i). So, in other words, the EKE decreases in Fig. 7h are caused by the effect of static stability acting directly on the eddies (Fig. 7j).

For the final experiment of this section, we show the response to a shallow heating over the polar regions, which is meant to emulate polar amplification under global warming. As found in both simple and comprehensive models (Butler et al. (2010); Peings and Magnusdottir (2014); Deser et al. (2015); Deser et al. (2016); McGraw and Barnes (2016); Wu and Smith (2016); Blackport and Kushner (2017)), the U response is an equatorward shift in the jet in both the QG (Fig. 8a) and PE (Fig. 3d) model. The majority of the U response is due to temperature gradients (Fig. 8b) but the static stability contribution is significant (Fig. 8c). Combining the temperature gradient and the advection of Θ by the $\overline{\omega}$ captures the full response almost completely (Fig. 8d). Therefore, the static stability change does not directly impact the eddies, instead the static stability changes meridional temperature gradients via the zonal-mean meridional circulation.

For EKE, the QG model underestimates the response on the equatorward flank of the jet (compare Fig. 8f and Fig. 3h), however, the location of the EKE dipole relative to the mean EKE is reproduced. The changes due to $\partial\theta/\partial y$ are strong and negative on the poleward flank of the mean EKE (Fig. 8g) and the changes due to static stability are a dipole centered on the mean EKE (Fig. 8h). As before, combining the temperature gradient and the advection of Θ by the $\overline{\omega}$ captures the full response almost completely (Fig. 8i). The only component of the static stability response that occurs directly via the effect of static stability on the eddies is a shallow, low-level increase in EKE (Fig. 8j). In the full response, however, this is offset by the $\partial\theta/\partial y$ response.

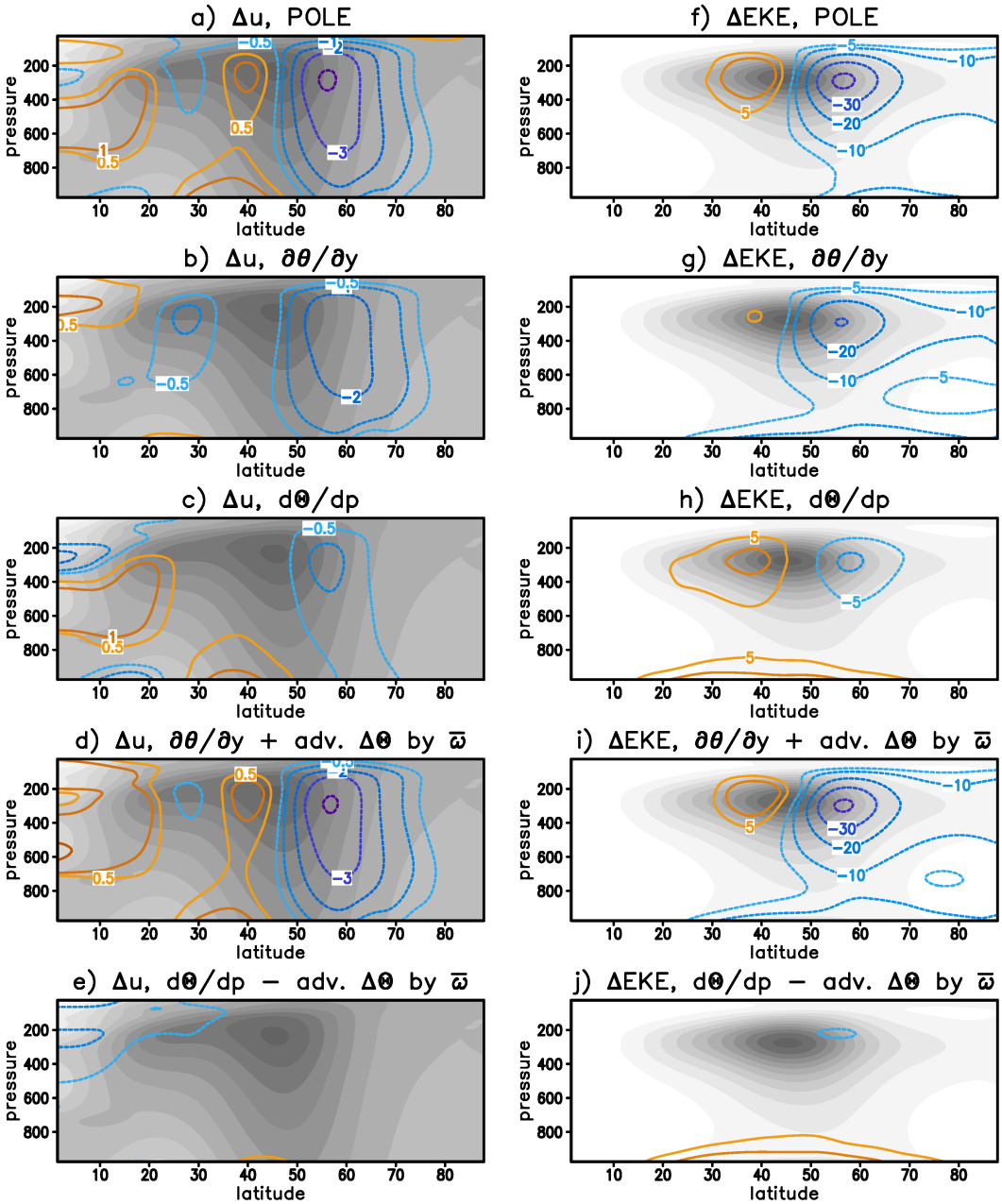

FIG. 8. As in Fig. 5 but for the POLE experiment (see Table 1).

d. Effect of Static Stability Alone

The QG model provides a unique opportunity to explore the role of static stability alone on the extratropical circulation. Therefore, in this section we explore the circulation response to static stability perturbations that are isolated in the vertical. The static stability perturbations have a

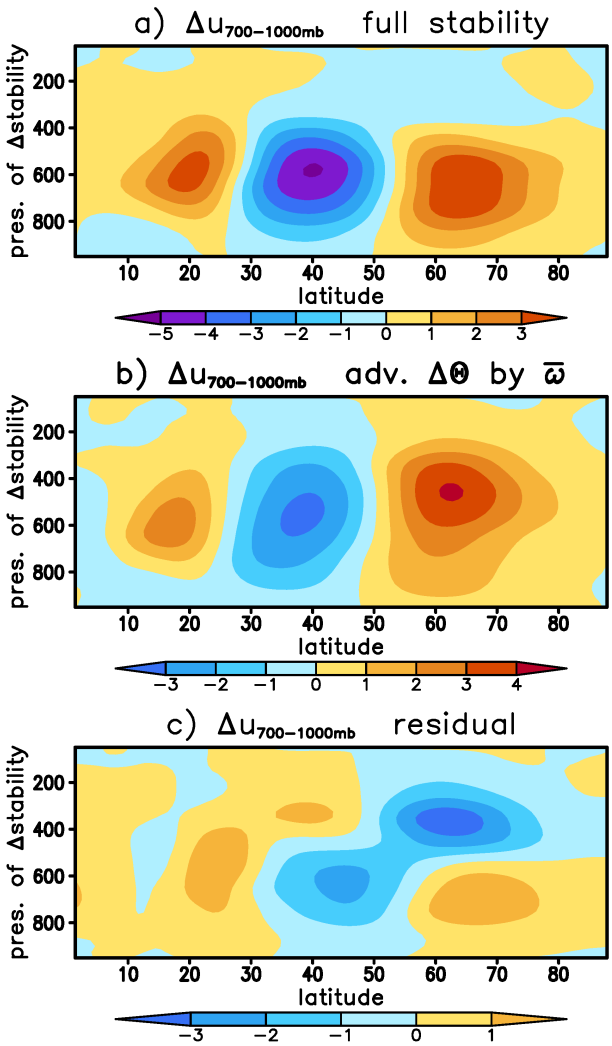

FIG. 9. a) Response of the zonal-mean zonal wind averaged vertically from 700-1000mb (x-axis) as a function of the pressure of the static stability perturbation (y-axis) in the QG model (m/s). b) Same as (a) but the static stability perturbation is replaced by a heat source equal to zonal-mean ω acting on the static stability perturbation (but the actual static stability is kept at control values). c) Residual: response in (a) minus (b).

Gaussian shape of width 100mb. There are ten experiments, which have perturbation locations ranging from 50mb to 950mb in increments of 100mb (see section 2d).

The U response averaged over the lower troposphere (700-1000mb) as a function of the pressure of the static stability perturbation is shown in Fig. 9a. Any increase in static stability below 400mb causes a poleward shift of the jet (mean jet location $\approx 49°$). Above this level the response is weak. The portion of the total U response caused by the mean control run $\overline{\omega}$ advecting the new reference

Θ is shown in Fig. 9b. As discussed previously, the latitude of the mean $\overline{\omega}$ (Fig. 2e) implies heating on the equatorward flank of the jet via vertical advection and cooling on the poleward flank. This strengthens the jet, which then causes a poleward shift via the mechanism in Kidston and Vallis (2012), Lorenz (2014a) and Lorenz (2014b) (see Introduction). The mean $\overline{\omega}$ advection response explains most of the total U response, although the location of the poleward center of action in Fig. 9b is displaced upward compared to Fig. 9a. The difference between the total and the mean $\overline{\omega}$ effect shows the impact of static stability directly on the eddies (Fig. 9c). This residual response peaks when the static stability perturbation is in the upper troposphere (350-450mb) and causes an equatorward shift in the jet. Nevertheless, this equatorward shift is completely offset by the zonal-mean $\overline{\omega}$ advection. At lower levels, static stability perturbations acting directly on the eddies cause a weak poleward shift of the jet that support the mean $\overline{\omega}$ advection response.

For EKE, horizontally uniform static stability changes primarily affect the vertical structure, therefore we show the global-mean EKE as a function of pressure (Fig. 10). Like Fig. 9, the vertical axis is the pressure of the static stability perturbation, but now the horizontal axis gives the vertical structure of the EKE response. Also, the line $y = x$ is shown for reference. For upper tropospheric static stability perturbations, the EKE decreases from just above the level of the imposed perturbation to the tropopause ($\approx$ 200mb) (Fig. 10a). For lower tropospheric static stability perturbations, the EKE decreases are weaker and are now below the level of the imposed perturbation and extend to the ground. The portion of the response that is a result of the advection of Θ by the mean $\overline{\omega}$ is shown in Fig. 10b. All perturbations lead to a small increase in EKE, which is consistent with the fact that the mean $\overline{\omega}$ is such that increased stability leads to increased meridional temperature gradients via advection. The residual EKE response (Fig. 10c), which shows the direct impact of stability on the eddies, is an enhanced version of Fig. 10a.

4. Conclusions

In this paper, a Primitive Equation (PE) model and Quasi-Geostrophic (QG) model are used to understand relative role of meridional temperature gradients and static stability on the response to thermal perturbations in a Held and Suarez (1994) GCM. We take advantage of the fact that the static stability in a QG model is externally prescribed and can therefore be altered independently of the meridional temperature gradient. First, the PE model is integrated to establish the static

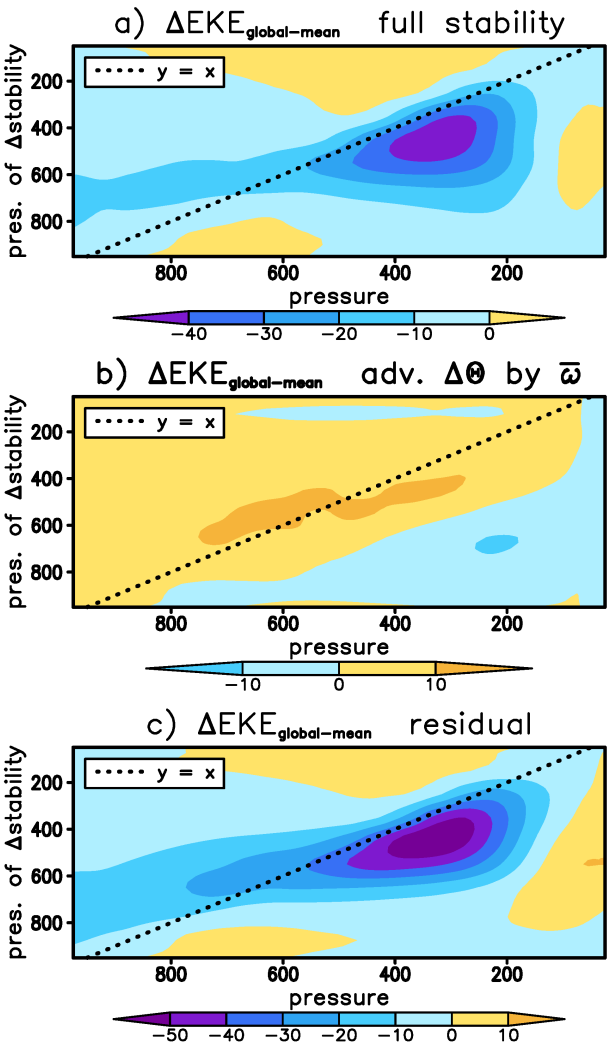

FIG. 10. a) Response of the global-mean EKE in pressure (x-axis) as a function of the pressure of the static stability perturbation (y-axis) in the QG model (m^2/s^2). The line $y = x$ is shown by the thick dotted line. b) Same as (a) but the static stability perturbation is replaced by a heat source equal to zonal-mean ω acting on the static stability perturbation (but the actual static stability is kept at control values). c) Residual: response in (a) minus (b).

stability of the control run and each thermal perturbation. Next, the QG model is integrated with the various combinations of control or perturbed Held and Suarez (1994) forcing, and control or perturbed static stability to understand the contribution of each to the full response. In this dry GCM, we focus on the extratropical response only.

Four thermal perturbations to Held and Suarez (1994) were explored in this study: 1) decreasing the stratosphere temperature, which increases the height of the tropopause, 2) increasing the static stability in the tropics, which is also associated with an increased pole-to-equator temperature gradient, 3) increasing the pole-to-equator temperature gradient while keeping the tropopause height at 45° fixed and 4) heating at low-levels over the pole to emulate polar amplification.

For zonal-mean zonal wind (U), changes in static stability are most important for the experiment with increased static stability in the tropics. For all other experiments, the meridional temperature gradient is more important. Except for increasing the tropopause height, the mechanism for the static stability response, however, does *not* directly involve the eddies, unlike the hypotheses in the literature (Frierson et al. (2006); Frierson (2008); Lu et al. (2008); Kidston et al. (2011)). Instead, the advection of the static stability perturbation by the zonal-mean vertical velocity accounts for almost all of the response to static stability. For increased static stability, advection by the Ferrel cell increases the meridional temperature gradient, which strengthens the jet via thermal wind. The stronger jet then shifts poleward, consistent with the theory of Kidston and Vallis (2012), Lorenz (2014a), and Lorenz (2014b) (see Introduction). The opposite response occurs for decreases in static stability. In other words, U is affected by the meridional temperature gradient both directly and indirectly (via static stability). As far as U changes are concerned, the static stability does not affect the eddy field directly. This picture is quite different from the literature and more work needs to be done to assess the role of this mechanism in moist models and observations.

For Eddy Kinetic Energy (EKE), the direct and indirect meridional temperature gradient changes explain essentially all of the response for polar amplification. For the other experiments the effect of static stability directly on the eddy field plays a non-negligible role, especially for the increase in tropopause height. In most cases considered here, however, the direct and indirect meridional temperature gradient effects are together more dominant. On the other hand, Yuval and Kaspi (2020) find that the relative role of static stability depends on the mean-state of the control climate and the structure of the temperature response. In future, work we will explore the sensitivity to the mean state in our model. We also systematically explored the impact of static stability perturbations localized in the vertical. The EKE response is largest above the perturbation when the perturbation is above 750mb, and the EKE response is largest below the perturbation when the perturbation is below 750mb.

Acknowledgments. This research was supported by NSF grant AGS-1557353.

Data availability statement. Please contact the author for the model code and data.

References

Barnes, E. A., and J. A. Screen, 2015: The impact of arctic warming on the midlatitude jet-stream: Can it? has it? will it? *Wiley Interdisciplinary Reviews: Climate Change*, **6 (3)**, 277–286.

Blackport, R., and P. J. Kushner, 2017: Isolating the atmospheric circulation response to arctic sea ice loss in the coupled climate system. *Journal of Climate*, **30 (6)**, 2163–2185.

Butler, A. H., D. W. J. Thompson, and R. Heikes, 2010: The steady-state atmospheric circulation response to climate change–like thermal forcings in a simple general circulation model. *Journal of Climate*, **23 (13)**, 3474–3496.

Chemke, R., and L. M. Polvani, 2019: Exploiting the abrupt 4×CO2 scenario to elucidate tropical expansion mechanisms. *Journal of Climate*, **32 (3)**, 859–875.

Chen, G., I. M. Held, and W. A. Robinson, 2007: Sensitivity of the latitude of the surface westerlies to surface friction. *J. Atmos. Sci.*, **64**, 2899–2915.

Deser, C., L. Sun, R. A. Tomas, and J. Screen, 2016: Does ocean coupling matter for the northern extratropical response to projected arctic sea ice loss? *Geophysical Research Letters*, **43 (5)**, 2149–2157.

Deser, C., R. A. Tomas, and L. Sun, 2015: The role of ocean–atmosphere coupling in the zonal-mean atmospheric response to arctic sea ice loss. *Journal of Climate*, **28 (6)**, 2168–2186.

Durran, D. R., 1991: The third-order Adams-Bashforth method: An attractive alternative to leapfrog time differencing. *Monthly Weather Review*, **119 (3)**, 702–720.

Durran, D. R., and P. N. Blossey, 2012: Implicit–explicit multistep methods for fast-wave–slow-wave problems. *Monthly Weather Review*, **140 (4)**, 1307–1325.

Frierson, D. M., I. M. Held, and P. Zurita-Gotor, 2006: A gray-radiation aquaplanet moist gcm. Part I: Static stability and eddy scale. *Journal of the atmospheric sciences*, **63 (10)**, 2548–2566.

Frierson, D. M. W., 2008: Midlatitude static stability in simple and comprehensive general circulation models. *Journal of the Atmospheric Sciences*, **65 (3)**, 1049–1062.

Frierson, D. M. W., J. Lu, and G. Chen, 2007: Width of the hadley cell in simple and comprehensive general circulation models. *Geophysical Research Letters*, **34 (18)**.

Gerber, E. P., and G. K. Vallis, 2007: Eddy-zonal flow interactions and the persistence of the zonal index. *J. Atmos. Sci.*, **64**, 3296–3311.

Haigh, J. D., M. Blackburn, and R. Day, 2005: The response of tropospheric circulation to perturbations in lower-stratospheric temperature. *J. Climate*, **18**, 3672–3685.

Harvey, B. J., L. C. Shaffrey, and T. J. Woollings, 2014: Equator-to-pole temperature differences and the extra-tropical storm track responses of the cmip5 climate models. *Climate Dynamics*, **43 (5)**, 1171–1182.

Harvey, B. J., L. C. Shaffrey, and T. J. Woollings, 2015: Deconstructing the climate change response of the northern hemisphere wintertime storm tracks. *Climate dynamics*, **45 (9)**, 2847–2860.

Haynes, P. H., and T. G. Shepherd, 1989: The importance of surface pressure changes in the response of the atmosphere to zonally-symmetric thermal and mechanical forcing. *Quarterly Journal of the Royal Meteorological Society*, **115 (490)**, 1181–1208.

Held, I. M., 1993: Large-scale dynamics and global warming. *Bulletin of the American Meteorological Society*, **74 (2)**, 228–242.

Held, I. M., and M. J. Suarez, 1994: A proposal for the intercomparison of the dynamical cores of atmospheric general circulation models. *Bulletin of the American Meteorological society*, **75 (10)**, 1825–1830.

Holland, M. M., and C. M. Bitz, 2003: Polar amplification of climate change in coupled models. *Climate dynamics*, **21 (3)**, 221–232.

Hoskins, B. J., and A. J. Simmons, 1975: A multi-layer spectral model and the semi-implicit method. *Quarterly Journal of the Royal Meteorological Society*, **101 (429)**, 637–655.

Kang, S. M., and J. Lu, 2012: Expansion of the hadley cell under global warming: Winter versus summer. *Journal of Climate*, **25 (24)**, 8387–8393.

Kang, S. M., and L. M. Polvani, 2011: The interannual relationship between the latitude of the eddy-driven jet and the edge of the hadley cell. *Journal of Climate*, **24 (2)**, 563–568.

Kidston, J., S. M. Dean, J. A. Renwick, and G. K. Vallis, 2010: A robust increase in the eddy length scale in the simulation of future climates. *Geophysical Research Letters*, **37 (3)**.

Kidston, J., and G. K. Vallis, 2012: The relationship between the speed and the latitude of an eddy-driven jet in a stirred barotropic model. *Journal of the Atmospheric Sciences*, **69 (11)**, 3251–3263.

Kidston, J., G. K. Vallis, S. M. Dean, and J. A. Renwick, 2011: Can the increase in the eddy length scale under global warming cause the poleward shift of the jet streams? *Journal of Climate*, **24 (14)**, 3764–3780.

Lorenz, D. J., 2014a: Understanding midlatitude jet variability and change using Rossby wave chromatography: Poleward-shifted jets in response to external forcing. *Journal of the Atmospheric Sciences*, **71 (7)**, 2370–2389.

Lorenz, D. J., 2014b: Understanding midlatitude jet variability and change using Rossby wave chromatography: Wave–mean flow interaction. *Journal of Atmospheric Sciences*, **71 (10)**, 3684–3705.

Lorenz, D. J., and E. T. DeWeaver, 2007: Tropopause height and zonal wind response to global warming in the IPCC scenario integrations. *J. Geophys. Res.*, **112**, D10 119.

Lorenz, E. N., 1960: Energy and numerical weather prediction. *Tellus*, **12 (4)**, 364–373.

Lu, J., G. Chen, and D. M. W. Frierson, 2008: Response of the zonal mean atmospheric circulation to El Niño versus global warming. *Journal of Climate*, **21 (22)**, 5835–5851.

Mbengue, C., and T. Schneider, 2017: Storm-track shifts under climate change: Toward a mechanistic understanding using baroclinic mean available potential energy. *Journal of the Atmospheric Sciences*, **74 (1)**, 93–110.

McGraw, M. C., and E. A. Barnes, 2016: Seasonal sensitivity of the eddy-driven jet to tropospheric heating in an idealized AGCM. *Journal of Climate*, **29 (14)**, 5223–5240.

O'Gorman, P. A., 2010: Understanding the varied response of the extratropical storm tracks to climate change. *Proceedings of the National Academy of Sciences*, **107 (45)**, 19 176–19 180.

Peings, Y., and G. Magnusdottir, 2014: Response of the wintertime northern hemisphere atmospheric circulation to current and projected arctic sea ice decline: A numerical study with CAM5. *Journal of Climate*, **27 (1)**, 244–264.

Rivier, L., R. Loft, and L. M. Polvani, 2002: An efficient spectral dynamical core for distributed memory computers. *Monthly Weather Review*, **130 (5)**, 1384–1396.

Simmons, A. J., and D. M. Burridge, 1981: An energy and angular-momentum conserving vertical finite-difference scheme and hybrid vertical coordinates. *Monthly Weather Review*, **109 (4)**, 758–766.

Whitaker, J. S., and A. F. Loughe, 1998: The relationship between ensemble spread and ensemble mean skill. *Monthly Weather Review*, **126 (12)**, 3292–3302.

White, A. A., 2002: A view of the equations of meteorological dynamics and various approximations. *Analytical Methods and Numerical Models*, **1**, 1–100.

Williams, G. P., 2006: Circulation sensitivity to tropopause height. *J. Atmos. Sci.*, **63**, 1954–1961.

Wu, Y., and K. L. Smith, 2016: Response of Northern Hemisphere midlatitude circulation to arctic amplification in a simple atmospheric general circulation model. *Journal of Climate*, **29 (6)**, 2041–2058.

Yin, J. H., 2005: A consistent poleward shift of the storm tracks in simulations of 21st century climate. *Geophysical Research Letters*, **32 (18)**.

Yuval, J., and Y. Kaspi, 2020: Eddy activity response to global warming–like temperature changes. *Journal of Climate*, **33 (4)**, 1381–1404.